Vo Van Tu Tai

QA Automation Engineer (Cypress · Detox · TypeScript · API & BDD Testing)

(+84) 869 500 573 • tutaivovan@gmail.com • linkedin.com/in/vovantutai • github.com/VoVanTuTai • tracking.nexus-ex.site

PROFESSIONAL SUMMARY

Information Systems Engineer (IUH, GPA 3.26/4.0 - Distinction) with a strong software engineering foundation in **JavaScript, TypeScript, and SQL**, combined with hands-on QA experience across **functional testing, API verification (Postman), and automated testing with Cypress**. Demonstrated experience at HomeNest Software formulating BDD test scenarios (Given-When-Then), conducting Jira defect triage, and verifying web/mobile staging builds against complex business logic. Solid comprehension of DOM architectures, asynchronous promises, and mobile gray-box testing concepts (Detox). Proactive, self-directed, and ready to contribute remotely to automated test suites.

CORE QA & TECHNICAL SKILLS

- **Test Automation & QA Methodologies:** Automated Web E2E Testing (Cypress — assertions, custom commands, network interception via cy.intercept), Mobile Gray-Box Testing Concepts (Detox — matchers, actions, synchronization), Functional & Regression Testing, Smoke Testing, Cross-Browser Testing.
- **Test Case Design & Defect Triage:** BDD Test Specifications (Given-When-Then acceptance criteria), Boundary Value Analysis, Equivalence Partitioning, Negative Testing, Jira Defect Tracking & Root Cause Analysis.
- **Programming & Scripting Languages:** TypeScript, JavaScript (ES6+, Async/Await, Promises, DOM Manipulation), SQL (PostgreSQL, MySQL — data verification queries, multi-table joins).
- **API Testing & Network Tools:** Postman (Automated test scripts, collection runners, environment variables, boundary payloads), RESTful APIs, OpenAPI 3.0 / JSON Schema validation, Chrome DevTools (Network, Console, Application inspection).
- **SDLC, DevOps & Collaboration:** Agile/Scrum, Git/GitHub (Pull requests, branch workflows), Docker Compose basics, CI/CD pipeline fundamentals; Technical English (Reading documentation, writing structured bug tickets and test documentation), Vietnamese (Native).

WORK EXPERIENCE

HomeNest Software — Ho Chi Minh City, Vietnam

Technical Bridge & QA Specialist (QA Testing · Business Analysis) — Mar 2026 – Sep 2026

Project 1: "Life Assistance" — Multi-Service On-Demand Platform & Dynamic Bidding

- **BDD Acceptance Criteria & API Contract Validation:** Translated complex multi-service business rules into clear Given-When-Then acceptance criteria; verified JSON contracts and OpenAPI 3.0 specifications in Postman to ensure schema compatibility prior to frontend integration.
- **Concurrency & Edge-Case State Testing:** Formulated rigorous edge-case test suites for two-way dynamic bidding (quote ↔ counter-bid with TTL expirations), verifying state locks and preventing race conditions or deadlock states during concurrent client actions.
- **GPS Telemetry & Event Ingestion Testing:** Verified event-triggered GPS coordinate logging logic across driver/rider states, checking payload sizes and boundary conditions to ensure zero event loss while reducing unneeded network overhead.
- **UI/UX Wireframe Audit & Defect Prevention:** Reviewed Figma wireframes against technical data models and validation constraints; identified UI inconsistencies and boundary gaps early, reducing change requests during active sprints.

Project 2: "VietPlus" — Classified Marketplace & Cross-Border Logistics

- **Functional & Smoke Testing Execution:** Executed comprehensive functional, regression, and smoke test suites across staging web and mobile webviews; verified multi-step listing forms, media uploads, and listing lifecycle state transitions.
- **Fulfillment & Dispute Workflow Verification:** Validated end-to-end order lifecycles including timeout cancellations, return disputes, and payment callback states, ensuring automated status progression complied with business specifications.
- **Jira Defect Tracking & Triage:** Logged clear, reproducible bug tickets with step-by-step instructions, screenshots, console logs, and payload traces; prioritized defects during sprint defect triage meetings with engineering leads.
- **Release Readiness & UAT Coordination:** Prepared mobile release verification checklists (App Store & Google Play submission requirements); facilitated User Acceptance Testing (UAT) sessions with client stakeholders, ensuring signed-off deliverables.

WORK EXPERIENCE (CONTINUED)

J&T Express Ho Chi Minh City, Vietnam

Data Verification & Operations Process Coordinator Sep 2025 – Feb 2026

- **SQL Data Auditing & Integrity Verification:** Formulated daily SQL audit queries (multi-table JOINs, aggregations, window functions) cross-checking physical hub intake manifests against database logs, ensuring 100% data consistency across thousands of parcels per shift.
- **Throughput Bottleneck Analysis:** Analyzed warehouse sorting telemetry to identify bottleneck stages; restructured operations into a one-way workflow (Inbound → Weigh → Scan → Dispatch), mitigating processing delays and sorting errors.
- **Standard Operating Procedures (SOP):** Authored standard operating procedure documentation for anomaly classification; trained operators in accurate system scanning, reducing human data entry discrepancies by 25%.

REPRESENTATIVE TECHNICAL & TESTED PROJECTS

Nexus Express — Order Management System (OMS) & Live AI Tracking Distributed Web System & E2E Testing

[Live Demo ↗] [GitHub ↗]

Cypress, Postman, NestJS, TypeScript, PostgreSQL, Camunda BPMN, React, Docker Sep 2025 – Feb 2026

- **Automated Web E2E Testing with Cypress:** Developed automated Cypress test suites validating core user journeys on tracking.nexus-ex.site (waybill search, invalid code error handling, real-time tracking timeline); leveraged cy.intercept() to stub API responses and assert UI states.
- **Idempotent API Contract Testing:** Formulated automated Postman test scripts validating OpenAPI 3.0 contracts; verified Idempotency Key logic under simulated network retries to confirm zero duplicate record creation in PostgreSQL.
- **BPMN Workflow State Verification:** Modeled and verified parcel lifecycle transitions against Camunda BPMN 2.0 gateway rules; checked edge cases across canceled, rerouted, and completed shipments.

Song Tai Shop & Tourist Accommodation — E-Commerce & Booking Platform [GitHub ↗] Web Application & Concurrency Testing

Postman, JavaScript, PHP MVC, MySQL (ACID), Webhook VNPay IPN, Docker Oct 2025 – Jan 2026

- **ACID Concurrency & Hold TTL Testing:** Formulated stress test scenarios to verify pessimistic locking (SELECT ... FOR UPDATE) and 15-minute temporary holds, confirming complete prevention of double-booking or overselling under simultaneous requests.
- **Payment Webhook & Checksum Validation:** Executed automated Postman tests verifying VNPay IPN webhooks; validated HMAC-SHA512 checksum signatures, idempotency handling, and accurate database order status updates.
- **Security & Input Validation:** Conducted security and boundary testing covering SQL injection resilience (PDO prepared statements), CSRF tokens, and session-based RBAC permissions.

Cab Booking & Real-Time Dispatch System — High-Concurrency Dispatch [GitHub ↗] Real-Time Dispatch Testing

Node.js, Redis (Distributed Lock SETNX), Leaflet, Postman, Docker Compose Dec 2025 – Jan 2026

- **High-Concurrency Race Condition Testing:** Simulated simultaneous driver acceptances using Postman runner collections; verified Redis distributed locks (SETNX with TTL) to guarantee exactly one driver is assigned per ride.
- **Geospatial Query & State Validation:** Verified Redis geospatial query outputs (GEOADD, GEORADIUS) against distance calculations; tested ride lifecycle state-machines (Requested → Accepted → Arrived → In-Trip → Completed).

EDUCATION & ACADEMIC BACKGROUND

Industrial University of Ho Chi Minh City (IUH) Ho Chi Minh City, Vietnam

Engineer in Information Systems (Completed all degree requirements · Available Full-Time) Sep 2022 – 2026

- **Cumulative GPA: 3.26 / 4.0** (Distinction / Xếp loại Giỏi).
- **Relevant Coursework:** Software Engineering & Testing, Object-Oriented Programming (OOP), Data Structures & Algorithms, Systems Analysis & Design (OOAD/UML), Database Systems & Advanced SQL (MySQL, PostgreSQL), SOA Architecture.